

The Utility Problem

With the trend of wide adaptation of new technologies in the distribution grid, like microgeneration (onsite PV), behind-the-meter energy storage and vehicle-to-grid (V2G), traditional approaches to grid management, assessment and forecasting need to be adapted to these new circumstances. A special focus should be dedicated to the adaptation of the load forecasting techniques since load forecasting (usually done on substation or feeder level) is a critical activity for every distribution utility.

One of the key concerns utilities have with these new technologies, as they pertain to load forecasting, is the possibility of a “hidden” or “ghost” load – a load on the system that is masked by local onsite generation (for example, PV generation or other DERs) but might rely on the grid to supply at any given moment (for the example, in the case of PV generation - if the sun is not shining in a given place). The utility’s system could be expected to supply this load, and yet its existence may be masked during normal operations. Utilities might also have concerns regarding their ability to accommodate an increase in loading due to these technologies in terms of grid capacity constraints.

Due to these reasons, for the utility to correctly forecast the amount of load, the knowledge of the consumption, which is offset by onsite generation, is crucial. Indicators of the ghost load, like total generation per circuit or % of the circuit’s load covered by renewables, are fundamental building blocks for correct load forecasting.

The Data Solutions and Methods

There are multiple methods to analytically approach this use case. These methods require different data sets to work with. First, it is important to determine if the goal of the analysis is to identify the presence of DERs behind the meter, which the utility did not know before or if the goal of the analysis is to go further and quantify the amount of hidden consumption which is offset by onsite generation. Further, it is important to take into account combination possibilities for behind-the-meter DERs in the service territory area. Some of the most common configurations are:

1. Solar PV only;
2. Solar PV in combination with battery;
3. Solar PV in combination with EV Charger;
4. Solar PV with battery and EV Charger.

Many of the utilities have records for the type and combination of behind-the-meter DERs, so it is possible to obtain the statistics for this type of data. These DERs and their combination can be approached as specific groups, which can be treated in different ways. It is worth noting that the presence of a behind-the-meter battery introduces additional complexity to the identification of ghost load.

Regardless of the details of the analytical approach, it is possible to summarize the required datasets for this type of analysis. If the utility wants to conduct ghost load analysis, these datasets will most likely include the following:

1. GIS Data or other datasets defining energy consumers/producers/DERs on the feeder or substation level;
2. Knowledge about the type of generation or DER. Possibly from the Client Information System (CIS);
3. Time series data in both quadrants (consumption and generation) for all utility meters from the examined feeder and/or substation;
4. Time series data from DERs like smart inverters, V2G chargers, batteries, etc;
5. Weather information, present and historical data.

It is important to mention that the Awesense Energy Transition Platform supports all the above-mentioned data sets and that developing the ghost load analysis is quite simplified since Awesense's Energy Data Model (EDM) interface provides numerous shortcuts for efficient development. After the data ingestion to the platform, all required datasets are centralized, synchronized and validated, which is essential for easier development of analytics.

The first analytical method looks at the presence of solar PV behind the meter as it should correlate with the weather during sunny and cloudy days. For this type of approach, it would also be beneficial to simulate the energy generation performance of the most common behind-the-meter PV configurations and compare the consumption against the cluster (cluster can be defined geospatially or by other segmentation attributes - both geospatial and attribute segmentation are supported and easily doable within the Awesense platform). After the simulation, the pattern can be identified and observed on pinpointed end consumers. After the confirmation, the amount of ghost load can be estimated for individual consumers and consequently aggregated onto the feeder or substation level.

The second analytical method assumes some level of "ground of the truth" where the utility would have data visibility into behind-the-meter DERs time-series (integrated data from the smart inverter, battery, etc.). This data, in combination with smart meter consumption data, would provide a form of training data set, and various machine learning (ML) methods can be applied afterwards to estimate the ghost loads. Awesense platform data access methods significantly simplify the retrieval of data, and the application of ML algorithms is convenient as analysts can focus on the performance of ML algorithms.

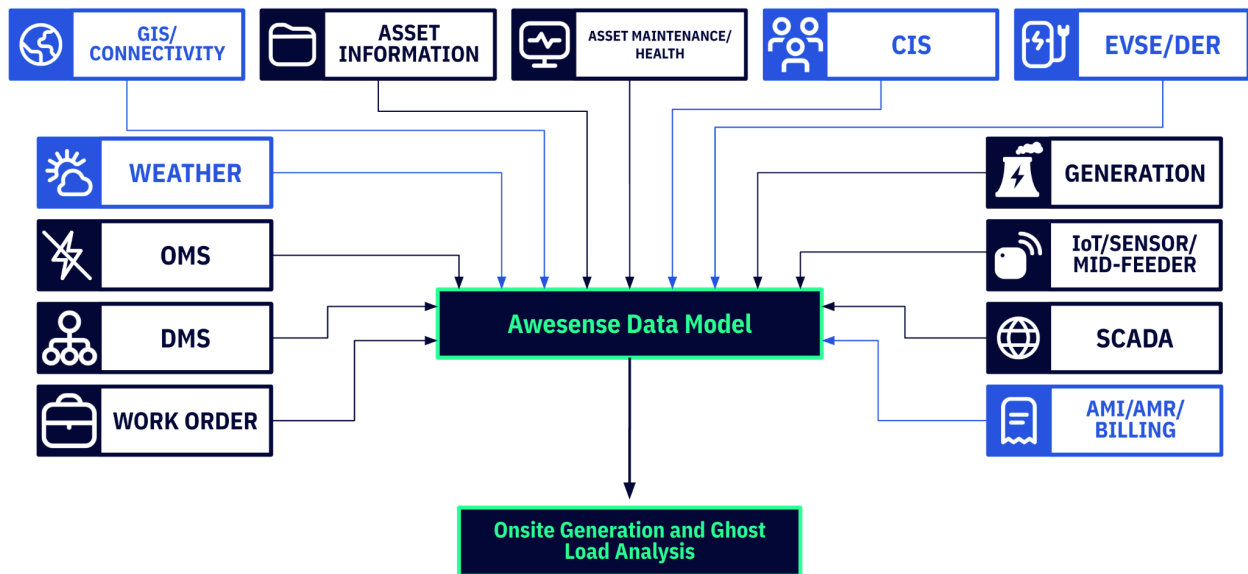
Onsite Generation and Ghost Load Analysis

[UC14-01] - Awesense Energy Transition Platform Use Case



Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. A GitHub link to the Jupyter notebook can be provided upon request.